



Maximizing weighted survival for coverage location model

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Abstract

Efficient deployment of emergency medical services (EMS) is critical for maximizing survival in life threatening situations while ensuring adequate coverage for non-critical calls. This study proposes a novel stochastic, survival probability and time dependent location allocation model for two ambulance types (advanced life support ALS and basic life support BLS) and two demand categories (critical and non-critical). A priority-based optimization framework ensures critical demand are served first, while improving non-critical coverage. To efficiently solve the NP-hard problem, a Memetic algorithm enhanced with iterated simulated annealing (MA-ISA) is developed, enabling optimal allocation of ambulances and demand assignments. Computational experiments in a real urban district demonstrate improved survival rates for critical calls and enhanced coverage for non-critical calls, offering a practical, dynamic tool for EMS resource planning under uncertainty.

Keywords Weighted survival function · Coverage location model · Memetic algorithm · Emergency medical services system · Coverage allocation model

1 Introduction

Emergency Medical Services (EMS) systems play a critical role in modern healthcare by providing rapid pre-hospital medical care and transportation for patients requiring urgent treatment. Timely response to emergency call is essential, as rapid medical intervention can significantly improve patient survival and reduce morbidity [1, 2]. A key operational indicator of EMS performance is response time that defined as the time required for an ambulance to arrive at the incident location after receiving an emergency call. Achieving short response times requires effective decisions regarding the location of ambulance stations and allocation of vehicles across service regions. This planning problem, commonly referred to as the EMS location allocation model which aims to

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determine appropriate station locations and ambulance assignments so that emergency demand can be served within predefined response time standards. One of primary factors affecting response time is travel time that depends on several uncertain factors such as traffic conditions, time of day, road characteristics and the weather. Consequently, travel time is inherently stochastic and this uncertainty should be considered when evaluating EMS performance [3, 4]. Ambulance planning decisions typically occur at multiple levels. Station location decisions are strategic, whereas allocation and redeployment decisions are tactical. These decisions are closely related and are often studied jointly in EMS model [5]. In practice, EMS system aims to meet predefined response time targets while efficiently allocating limited ambulance resources [6]. As a result, many studies evaluate EMS performance using coverage-based metrics, which measure the amount of demand that can be served within a specified time threshold [2, 6]. Although coverage metrics are widely used, they do not fully capture the effect of response time on patient outcomes. For life threatening emergencies, patient survival probability often decreases rapidly as response time increases. Therefore, several studies have proposed incorporating survival-based performance measures into EMS location models [7]. Considering multiple EMS performance metrics can improve the realism of EMS planning models, particularly in systems that served patients with different levels of urgency [8, 9]. Another important issue in EMS planning is ambulance availability. Several studies therefore estimate the probability that ambulance is busy when calls arrive [10, 11]. In addition, emergency demand typically varies over time, with fluctuations across different periods of the day of the week. Thus, ignoring this temporal variation may lead to inaccurate estimates of the number of ambulances required to maintain service performance [4, 12]. Ambulance location and allocation model are generally classified as NP-hard optimization problems, which makes solving large scale instances computationally difficult [13, 14]. Consequently, many heuristic and meta-heuristic methods have been proposed to efficiently solve EMS planning problem [15, 16].

Motivated by these challenges, this study proposes a stochastic EMS survival location allocation model that determine the optimal deployment of two ambulance types: advanced life support (ALS) units and basic life support (BLS) units. The model incorporates stochastic travel times modeled using a Weibull distribution, survival probability estimation through a logistic survival function, and ambulance availability estimated using an approximate hypercube queuing model. To reflect operational priorities in EMS systems, the proposed model adopts a priority optimization framework in which the survival probability of critical patients is maximized first, followed by improvement of coverage for non-critical calls. Because the resulting problem is non-linear, stochastic and computationally complex, a hybrid Memetic algorithm combined with an incremental search algorithm (MA-ISA) is developed to efficiently solve large scale instances. The proposed approach is evaluated utilizing real EMS demand data from Al-Karakh district in Baghdad, Iraq, which presents a large urban service area with heterogeneous emergency demand [17]. The main contributions of this study are summarized as follows:

- Development of a stochastic EMS survival location allocation model integrating survival and coverage-based performance measures.

- Incorporation of survival probability modeling, stochastic travel time, and ambulance availability utilizing an approximation hypercube queuing model.
- Adoption of a priority optimization framework prioritizing survival of critical demand before improving non-critical demand.
- Development of a hybrid MA-ISA meta-heuristic algorithm for solving large scale EMS planning problems.
- Comprehensive computational evaluation utilizing real demand dataset as well as simulated instance of varying sizes furthermore benchmark experiments comparing the propose algorithm with an exact solver and an alternative EMS meta-heuristic.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on EMS location models. Section 3 presents the proposed model, while Sect. 4 describes the solution methodology. Sections 5 reports computational experiments. Section 6 provides the computational results and accompanying commentary. Finally, Sect. 7 ultimately delineates the results and identifies avenues for future investigation.

2 literature review

EMS systems require efficient planning to ensure rapid response and high service reliability. Over the past decade, numerous studies have investigated the design, location and deployment of emergency medical resources using optimization and simulation approaches. Comprehensive reviews of EMS logistic and operational challenges are provided in [12, 18–20]. Early research on emergency facility location introduced several classical models. The Location Set Coverage Model LSCM proposed by Toregas et al. [21] determines the minimum number of facilities required to cover all demand nodes within a specified response time. Church and ReVelle [22] later introduced the Maximal Expected Covering Location Problem MEXCLP, which aims to maximize the amount of demand covered by limited number of facilities. To account for ambulance busy probabilities Daskin [23] developed the Maximal Expected Covering Location Problem MEXCLP. Another important analytical framework is hypercube queuing model proposed by Larson [24], which estimates ambulance availability and system performance in stochastic environments. Several extensions have been developed to improve the reliability and efficiency of EMS coverage models. Subsequent studies extended these approaches to include backup coverage strategies and improved estimation of ambulance busy probabilities [25, 26].

On the other hand, Hogan and ReVelle [27] introduced backup coverage concepts to ensure service availability when primary ambulances are busy. Mandell [28] provided a coverage model for a dual-tier EMS system in which ALS and BLS are capable of addressing all emergency demand. A territory is deemed covered if an ALS unit is accessible for assistance inside a certain response time. The Maximum Covering Location Problem MCLP Proposed by ReVelle and Hogan [29] considers the probability that at least one ambulance is available to serve a demand point. Further improvements include probabilistic coverage models [30] and queuing based availability models such

as the Queuing Maximal Availability Location Problem QMALP developed by Marianov and Reville [31]. To better represent real world EMS operations, researchers have also incorporated dynamic and time dependent features into ambulance location models. Rajagopalan et al. [32] proposed a multi period set covering model for the dynamic redeployment of ambulances. Schmid and Doerner [33] introduced ambulance relocation models that consider time dependent travel times. Later, Saydam et al. [34] developed dynamic redeployment coverage location models that improve EMS responsiveness in changing demand environments. In addition, Yoon and Albert [35] proposed queue-based coverage models considering dispatch priorities.

Because rapid response is strongly related to patient survival, several studies have incorporated survival-based objectives into ambulance location models. Erkut et al. [7] developed an ambulance location model that directly maximizes patient survival based on response time distributions. Knight et al. [36] extended this approach by incorporating heterogeneous survival outcomes for different types of emergencies. Recent research has also examined heterogeneous ambulance fleets and hierarchical EMS systems. Leknes et al. [10] studied strategic ambulance location problems in heterogeneous geographic regions. Nelas and Dias [37] proposed a hierarchical location model that considers programming model for locating and dispatching multiple types of ambulances. Due to the computational complexity of EMS location problems, meta-heuristic algorithms have been widely applied. Rajagopalan et al. [18] investigated meta-heuristic approaches for probabilistic location model. Doerner et al. [38] developed heuristic solution methods for extended ambulance location problems, while Zhen et al. [39] proposed simulation optimization framework for ambulance deployment and redeployment decisions. Akdogan et al. [40] combined approximate queuing model with meta-heuristic algorithm for locating emergency units. In previous studies, Naji et al. [41] proposed a two server dynamic coverage location model under stochastic travel time conditions. Later, Naji et al. [42] developed a multi objective facility coverage model for emergency medical service planning in large scale events. More recently, evolutionary and Memetic algorithms have been applied to ambulance location problem [43], building on the Memetic algorithm framework introduced in [44] and [45].

Despite these advancements, many existing models focus either on coverage maximization or simplified service structures. Few studies integrate survival based objective, stochastic travel times and advanced meta-heuristic optimization within a unified framework. Therefore, this study proposes an enhanced ambulance location optimization approach using a Memetic algorithm to improve EMS performance and decision making efficiency.

3 EMS survival location-allocation model

This section presents a stochastic, time-dependent EMS location-allocation model that determines the optimal deployment of two ambulance types (ALS and BLS) in response to two categories of incoming demand (critical and non-critical calls). The proposed model aims to: maximize survival probability of critical calls served by either ALS or BLS ambulances as well as maximize the coverage of non-critical calls served

by BLS ambulances while maintaining the optimal survival level. The suggested model explicitly incorporates: stochastic travel times using Weibull distribution, survival probability of critical calls using logistic survival function and ambulance availability using an approximation hypercube queuing model. Unlike traditional EMS deployment models that combine several performance metrics using weighted objectives, the proposed formulation adopts a priority optimization approach. In this framework:

- Primary objective: maximize survival probability of critical calls.
- Secondary objective: maximize coverage of non-critical calls, subject to optimal survival.

This structure reflects the operational priorities of EMS system, where life threatening emergencies must be addressed before service coverage considerations.

3.1 Assumptions

The following presumptions are adopted to formulate the proposed model.

- The EMS system operates with two types of ambulances: the ALS ambulances, staffed by paramedics and BLS ambulances, staffed by the Emergency Medical Technicians (EMTs).
- ALS ambulances are primarily responsible for critical calls, while BLS ambulances serve non-critical calls and used as a "backup" for critical calls when ALS units are unavailable.
- Emergency calls (critical, non-critical) arriving from demand zone i at time t follow a Poisson arrival process with arrival rate λ_{it}^c , λ_{it}^n , for critical and non-critical calls respectively.
- Travel time T_{ij}^k between location j and node i for ambulance type $k \in \{A, B\}$ follows a Weibull distribution with shape parameter $\alpha_{ij}^k > 0$ and scale parameter $\beta_{ij}^k > 0$ [42]. Thus, a call is considered covered if response time is within a predefined threshold s^A, r^B for ALS / BLS, respectively. The set of ALS / BLS within the critical / non-critical node i time standard thresholds TH^A, TH^B describes as follows:
 $TH^A = \{j | \beta_{ij}(-\ln(1 - \gamma))^{1/\alpha_{ij}} \leq s^A\}$, $TH^B = \{j | \beta_{ij}(-\ln(1 - \gamma))^{1/\alpha_{ij}} \leq r^B\}$.
- Ambulances ALS/BLS operate independently and ambulance availability probabilities are estimated use to approximate hypercube queuing system. According to [41], this approach allows estimation of busy probabilities and correction factors for ALS/ABLS ambulances at time period t .

3.2 Notation

3.2.1 Sets and indices

N : Set of demand node i .

J : Set of eligible ambulance facility j (ALS / BLS).

T : Set of time intervals t .

$k \in \{A, B\}$: Ambulance type (ALS, BLS).

3.2.2 Parameters

K_t^A, K_t^B : Number of available ALS / BLS ambulances at time t , respectively.

ρ_t^A, ρ_t^B : Average busy probability of ALS and BLS ambulances at a time t , respectively.

ρ_{jt}^A, ρ_{jt}^B : Busy probability of the j th ALS and BLS ambulance at time t , respectively.

P^A, P^B : Probability of all ambulances of type ALS and BLS are busy, respectively.

P_0^A, P_0^B : Probability of all ambulances of type ALS and BLS are not busy, respectively.

$\lambda_{it}^c, \lambda_{it}^n$: Arrival rate of critical and non-critical from node i at time t , respectively.

λ_t^c, λ_t^n : Total number of critical and non-critical calls from node i that arrival at time t , respectively.

τ_i^A, τ_i^B : Expected service time of the ALS / BLS at node i , respectively.

τ^A, τ^B : Average service time of the ALS / BLS ambulances in the system, respectively.

h_{it}^c, h_{it}^n : Fraction of critical / non-critical calls served by ALS/BLS ambulance at time t , respectively.

C_t^A, C_t^B : Minimum cover requirements for ALS / BLS at time t , respectively.

α_i^c, α_i^n : Required reliability for critical / non-critical calls at node i , respectively.

$Q^A(K_t^A, \rho_t^A, j^A), Q^B(K_t^B, \rho_t^B, j^B)$: Correction factors of ALS/BLS, respectively derived from an approximation hypercube $M/M/m/loss$ system.

W_{it}^c, W_{it}^n : Performance weight related to critical / non-critical call arrival from node i at duration t if they are covered by eligible facility (ALS/BLS), respectively.

3.2.3 Decision variables

$X_{ijt}^A, X_{ijt}^B = 1$ if that the j th ALS / BLS ambulance is at node i at time t , 0, otherwise, respectively.

$Y_{it}^A, Y_{it}^B = 1$ if that at least one ALS / BLS is within threshold TH^A, TH^B of node i at time t , 0, otherwise, respectively.

a_{ijt}^A, a_{ijt}^B : 1 if node i falls inside the threshold TH^A and TH^B of j th ALS / BLS at time t , 0, otherwise, respectively.

3.3 Objective function

The primary aim of the proposed EMS model is to maximize the operational performance of the EMS system by optimally locating ALS and BLS ambulances across the planning horizon t . The performance of each ambulance type depends on the stochastic travel time T_{ij}^k from eligible facility j to node i for ambulance type $k \in \{A, B\}$. For

critical calls, the system performance is evaluated using a logistic survival function [7]. The survival probability of the critical calls associated with response time t is expressed as: $S^c(t) = 1/(1 + e^{-0.679+0.262t})$. Since the response time is stochastic, the expected survival probability is obtained by integrating the survival function over the Weibull travel time distribution:

$$\bar{S}_{ij}^k = \int_0^{\infty} S^c(t) f_{ij}^k(t) dt$$

where, $f_{ij}^k(t) = \frac{\alpha_{ij}}{\beta_{ij}} \left(\frac{t}{\beta_{ij}}\right)^{\alpha_{ij}-1} e^{-(t/\beta_{ij})^{\alpha_{ij}}}$ denotes a Weibull probability density function of travel time. Since the ideal station placement is independent of the survival function's parameters, any comparable monotonically declining function maybe utilized [18].

For non-critical calls, a binary coverage metric is used:

$$S^n(t) = \begin{cases} 1, & \text{if response time} \leq r^B \\ 0, & \text{otherwise} \end{cases}$$

As the ambulances (ALS/BLS) may be busy responding to other calls, the survival and coverage probabilities must be adjusted by the available probabilities. Thus, the probability that ALS, BLS ambulances are available define as: $1 - \rho_{jt}^A$, $1 - \rho_{jt}^B$, respectively. These probabilities are estimated using the approximation hypercube model are incorporated into the performance measures to obtain the effective survival and coverage probabilities. To reflect the operational priorities of the EMS system, the proposed model adopts a priority optimization framework. In this framework, two performance measures are considered: the survival probability of critical calls is treated as the primary objective, while the coverage of the non-critical calls is treated as a secondary objective. Since EMS systems prioritize life threatening emergencies, the primary objective function of proposed model maximizes an expected survival number of critical calls served by ALS or BLS ambulances before improving the coverage for non-critical calls served by BLS ambulances. The critical survival primary objective function (when critical calls served by ALS (preferred) or by BLS if ALS is unavailable) is defined as:

$$\text{Maximize } Z_1 \sum_{t=1}^T \sum_{i=1}^N \left[W_{it}^c (1 - \rho_{jt}^A) \bar{S}_{ij}^A y_{it}^A + W_{it}^c (1 - \rho_{jt}^B) \bar{S}_{ij}^A y_{it}^B \right]$$

The second objective function that maximizes an expected coverage of non-critical calls served by BLS ambulances while maintaining the optimal survival level is defined as:

$$\text{Maximize } Z_2 \sum_{t=1}^T \sum_{i=1}^N W_{it}^n (1 - \rho_{jt}^B) \bar{S}_{ij}^B y_{it}^B$$

The final priority objective expressed as: step 1 *Maximize* Z_1 and step 2 *Maximize* Z_2 s.t. $Z_1 = Z_1^*$. Priority objective ensures that the second objective (improvements in non-critical service coverage) are pursued only after achieving the primary objective (survival performance for critical call) has been maximized.

3.4 Constraints

The model that has been provided has a constraint that ensures an essential demand from node i is only considered if it can be handled by an ambulance equipped with advanced / basic life support (ALS/BLS) with a reliability α_i^c :

$$\left\{ 1 - \prod_{j=1}^{K_t^A} \rho_{jt}^{A \sum_{i=1}^N a_{ijt}^A X_{ijt}^A} Q^A \left(K_t^A, \rho_t^A, \sum_{i=1}^N \sum_{j=1}^{K_t^A} a_{ijt}^A X_{ijt}^A - 1 \right) + 1 - \prod_{j=1}^{K_t^B} \rho_{jt}^{B \sum_{i=1}^N a_{ijt}^B X_{ijt}^B} Q^B \left(K_t^B, \rho_t^B, \sum_{i=1}^N \sum_{j=1}^{K_t^B} a_{ijt}^B X_{ijt}^B - 1 \right) \right\} \geq \alpha_i^c, \forall i, t \quad (1)$$

Similarly, the constraint that ensures a non-essential demand from node i is handled by a BLS with a reliability α_i^n might be implemented in the following manner:

$$\left\{ 1 - \prod_{j=1}^{K_t^B} \rho_{jt}^{B \sum_{i=1}^N a_{ijt}^B X_{ijt}^B} Q^B \left(K_t^B, \rho_t^B, \sum_{i=1}^N \sum_{j=1}^{K_t^B} a_{ijt}^B X_{ijt}^B - 1 \right) \right\} \geq \alpha_i^n, \forall i, t \quad (2)$$

where, $\left(1 - \prod_{j=1}^{K_t^A} \rho_{jt}^{A \sum_{i=1}^N a_{ijt}^A X_{ijt}^A} \right)$, $\left(1 - \prod_{j=1}^{K_t^B} \rho_{jt}^{B \sum_{i=1}^N a_{ijt}^B X_{ijt}^B} \right)$ are represent probabilities

that at least one of the ALS/BLS ambulances are available within TH^A , TH^B , respectively. $Q^A \left(K_t^A, \rho_t^A, \sum_{i=1}^N \sum_{j=1}^{K_t^A} a_{ijt}^A X_{ijt}^A - 1 \right)$, $Q^B \left(K_t^B, \rho_t^B, \sum_{i=1}^N \sum_{j=1}^{K_t^B} a_{ijt}^B X_{ijt}^B - 1 \right)$ are correction factors of the ALS/BLS ambulances, respectively.

Often, due to budget restraints, the quantity of available ALS/BLS are limited. Expressions $\sum_{j=1}^{K_t^A} X_{ijt}^A$, $\sum_{j=1}^{K_t^B} X_{ijt}^B$ describe how many ALS/BLS present in the system, respectively. Hence, to limit the number of available ALS/BLS located we have the following constraint:

$$\sum_{j=1}^{K_t^A} X_{ijt}^A \leq K_t^A, \sum_{i=1}^{K_t^B} X_{ijt}^B \leq K_t^B \forall t \quad (3)$$

Hence, the constraints to guarantee that all calls (both critical/non-critical) are reliably attended by ALS or BLS with cover reliability α_i^c , α_i^n , respectively cannot be lower than the minimum coverage requirements C_t^A , C_t^B are formulated as follows:

$$\sum_{i=1}^N h_{it}^A y_{it}^A \geq C_t^A, \quad \sum_{i=1}^N h_{it}^B y_{it}^B \geq C_t^B \quad \forall t \quad (4)$$

The average probability of ALS ambulances busyness at time t is expressed as follows:

$$\rho_t^A = 1 - \left(\sum_{j=1}^{K_t^A} \rho_{j,t}^A \right) / (K_t^A) (\rho_{t,\text{initial}}^A) \quad (5)$$

where, $\rho_{t,\text{initial}}^A = (\lambda_t^A)(\tau_i^A)/K_t^A$ is the initial busy probability of the ALS ambulances at time and $\rho_{j,t}^A$ is approximation for the ALS busy probabilities:

$$\rho_{j,t}^A = (V_j^A) / (1 + V_j^A)$$

$$V_j^A = \sum_{j^A=0}^{K_t^A-1} \sum_{\alpha_{il}^A=j} \lambda_t^A \tau_i^A Q^A(K_t^A, \rho_t^A, j^A) \prod_{l=1}^{k-1} \rho_{t,\alpha_{il}^A}^A$$

The arrival rate of calls (critical/ non- critical) at time t denoted as λ_t^A , λ_t^B and τ_i^A , τ_i^B represent the expected service time for the ALS/ BLS ambulance at node i . Additionally, K_t^A , K_t^B indicates the quantity of ALS/BLS at time t . The correction factor of the ALS represented as $Q^A(K_t^A, \rho_t^A, j^A)$, is computed using an approximation hypercube. This method considers the number of the ALS ambulances at time t , the average busy probability of the ALS fleet at that time t , and the $(j^A)^{th}$ preferred ALS ambulance that provides coverage for the critical call ($j^A = 0, 1, \dots, K_t^A - 1$). Also, α_{il}^A refers to the index of ALS or back-up BLS ambulances that are occupying the l^{th} preference position for critical demand arising from node i . Similarly, the average active probability of the BLS ambulances at time t is expressed as follows:

$$\rho_t^B = 1 - \left(\sum_{j=1}^{K_t^B} \rho_{j,t}^B \right) / (K_t^B) (\rho_{t,\text{initial}}^B) \quad (6)$$

where, $\rho_{t,\text{initial}}^B = (\lambda_t^A + \lambda_t^B)(\tau_i^B)/K_t^B$ is the initial busy probability of the BLS ambulances at time t , $\rho_{j,t}^B$ is approximation for the BLS busy probabilities:

$$\rho_{j,t}^B = (V_j^B) / (1 + V_j^B)$$

$$V_j^B = \sum_{j^B=0}^{K_t^B-1} \sum_{\alpha_{it}^B=j} \lambda_t^B \tau_i^B Q^B(K_t^B, \rho_t^B, j^B) \prod_{l=1}^{k-1} \rho_t^B \alpha_{il}^B$$

Fraction constraint of critical contacts originating from node i that are handled by an ALS or a BLS at time t is articulated as follows:

$$h_{i,t}^A = \sum_{j^A=0}^{K_t^A-1} Q^A(K_t^A, \rho_t^A, j^A) (1 - \rho_t^A) \prod_{l=1}^{k-1} \rho_t^A \alpha_{il}^A \quad (7)$$

Likewise, the fraction constraint of non-critical calls from node i that are served by BLS ambulance at time t is given as follows:

$$h_{i,t}^B = \sum_{j^B=0}^{K_t^B-1} Q^B(K_t^B, \rho_t^B, j^B) (1 - \rho_t^B) \prod_{l=1}^{k-1} \rho_t^B \alpha_{il}^B \quad (8)$$

The constraints that force coverage variables to be binary (i.e. 0, 1) is formulated as follows:

$$X_{ijt}^A, X_{ijt}^B, Y_{it}^A, Y_{it}^B = \{0, 1\} \forall i, j, t \quad (9)$$

The following condition confirms that the set of ALS / BLS units located within the designated time standard threshold, denoted by TH^A , TH^B for critical and non-critical demand node i :

$$TH^A = \left\{ j | \beta_{ij} (-\ln(1 - \gamma))^{1/\alpha_{ij}} \leq s^A \right\}, TH^B = \left\{ j | \beta_{ij} (-\ln(1 - \gamma))^{1/\alpha_{ij}} \leq r^B \right\} \quad (10)$$

4 Solution methodology

The proposed EMS survival model is a stochastic, time-dependent location-allocation problem with two type ambulances (ALS/BLS) and two categories demand (critical and non-critical). This problem is combinatorial, non-linear, and NP-hard [12] due to:

1. Discrete placement decisions for multiple ambulance units,
2. Stochastic travels times modeled by Weibull distributions,
3. Survival probability computations via logistic function,
4. Ambulance availability estimated using an approximation queuing hypercube model, and
5. Priority objectives maximizing survival for critical calls before maximizing non-critical coverage.

Given the size and stochastic nature of the problem, exact methods (e.g., integer linear programming) become computationally intractable for realistic networks. The complexity of the issue is increased of the rapid development of the solution space, as well as the addition of individual station level busy likelihood and limits associated to

relocation. As the quantity of demand zones, potential sites, and time periods increases, the amount of processing time that is required to arrive at a solution that is suitable and feasible also increases. Consequently, a meta-heuristic search approach is employed in lieu of integer linear programming to address the EMS models that necessitate parameter calculations during runtime. The meta-heuristic search approach facilitates the computation of parameters in real-time and guarantees "good" or "near-optimal" solutions with little processing time [46]. A fundamental attribute of meta-heuristic strategies is their capacity to evade local optima by employing a strategy that facilitates departure from local optima and conducts an extensive search of a potential region [34, 47]. Numerous publications [48, 49] have employed diverse meta-heuristic methodologies to address ambulance placement issues; we cite Talbi, E. G. [50] for a recent complete assessment of these meta-heuristic strategies. This study employs a Memetic Algorithm (MA) technique that enhances the Incremental Search Algorithm (ISA) by including a local search routine. MA is a population-centric meta-heuristic that integrates natural adaptation with individual learning obtained by the population members [43]. MA accelerates the identification of global optimal conditions and prevents convergence that occurs too sooner by harmonizing the examination of search space through ISA while facilitating the utilization of the existing neighborhood via local search techniques. ISA is significant meta-heuristic search technique that incorporates reactive Tabu search (RTS) algorithm, designed to effectively address dynamic issues characterized by temporal changes, such as environmental updates or shifts in target position. Meta-heuristic algorithms have been effectively utilized in addressing diverse combinatorial challenges, including scheduling [51], vehicle routing [52], assignment difficulties [53], networking design for supply chains [54], as well as distribution of locations [55, 56]. MA has been employed in addressing healthcare planning challenges, including the distribution of ambulances [57] and regarding the planning of healthcare at home [58]. Multi-agent systems often employ binary or integer encoding for the purpose of solution description in the literature. There have been real-coded Memetic algorithms proposed for addressing optimization issues involving continuous variables. Similar to the distribution of locations issue which integrates strategic location facility decisions [45] and tactical routing considerations [59], an ambulance location-allocation problem amalgamates both strategically and tactically decision-making levels.

4.1 Suggested algorithm

To address the suggested model outlined in the Sect. 3, the following algorithm was constructed. The MA enhanced with local search using the ISA is adopted to ascertain the optima placement of stations, the allocation of call zones to these stations, and the distribution of two categories of units (both ALS and BLS) which respond to essential demand, as well as BLS which address both critical and non-critical calls, by maximizing the survival probability of critical calls and maximizing the coverage requirements for non-critical calls. Henceforth, the improved MA will be termed as the MA-ISA search algorithm. The proposed search MA-ISA algorithm has two parts.

The following sections outline the principal stages involved in the implementation of the proposed algorithm:

4.1.1 Module 1: Initial solution generation

The first module generates a feasible initial solution and evaluates the coverage limitations using Jarvis' hypercube approximation approach. This module account for:

- Stochastic travel times and survival probabilities for critical calls,
- Ambulance availability probabilities for both ALS and BLS,
- Coverage requirements for non-critical calls, and
- Evaluating survival probabilities.

The initial number of ALS/BLS units at each time period t known as one-dimensional array [53]. The size of the array is given as $(K_t^A + 1, K_t^B + 1)$, where K_t^A, K_t^B represent the available of ALS/BLS at time interval t . During the first iteration at time t , the preliminary fleet sizes were determined by using the following equations: $K_t^A = \lambda_t^c / \rho_t^A \mu_t^A$, $K_t^B = \lambda_t^n / \rho_t^B \mu_t^B$. At the beginning of each array is the coverage value and the locations of ALS/BLS at time interval 1, and this process is repeated for all of the time intervals. A random number generator was used to create the locations of the ambulances in order to determine the preliminary fleet size at time t . using recomputed travel times and hypercube corrections, the coverage probabilities are calculated for each node and time periods. These values represent whether a node is within response time threshold of a particular ambulance. Preliminary solutions are checked against minimum coverage requirements and ambulance availability constraints, ensuring all initial chromosomes are feasible for the MA. The initial solutions for MA should be established based on these findings.

4.1.2 Module 2: Employing a memetic algorithm with ISA as a local search

The second module refines the initial solutions using an MA algorithm combined with ISA as a local search mechanism. This hybrid approach balances global exploration and local exploitation, handling both discrete ambulance location and continues demand allocations. Additionally, a selection technique was utilized in order to pick parent chromosomes that are susceptible to crossover and mutation to create offspring. For the purpose of accommodating both continuous and discrete variables inside the model, a mixed coded representation was developed. In this representation, each chromosome that depicts a solution is expressed as combination of two vectors, namely facility vector and call vector. There are two kinds of facility vectors: facility vectors F^A, F^B represent the ALB/BLS ambulances that are assigned to each demand zone, which is a discrete variable. The call vectors $C^{critical}, C^{non-critical}$ represent the fraction of demand from zone that is being served by the corresponding ALS/BLS units, respectively, which is a continuous variable. In accordance with the total demand that is allocated to a particular station, the distribution of ALS/BLS to that station is an important consideration. The selection phase is an essential part of the MA process. There are a variety of selection methods accessible, such as the selection of tournament,

the selection of the roulette wheel selection, the selection of rank, and the selection of the Boltzmann [60]. In this particular investigation, the technique that was applied was drawn from the ranking selection strategy that was given by Reeves [61]. The first step is to rank-based selection priorities feasible solutions with higher expected survival. The probability of selection for chromosome k is: $P(k) = 2k/M(M+1)$ where, k represents the position of the chromosome within the ordered list of objective values, and M denotes the size of population. This method is beneficial due to its simplicity of implementation and its focus on solutions that exhibit higher function values and fewer violations of constraint. The solution with the optimal fitness value has a probability M times higher than that of the lowest-ranked solution and twice that of the median value. The two selected chromosomes are recombined to produce new offspring during the crossover phase. During the crossover phase, we apply a single-point crossover (BLX) on the facility vector and the BLX- α crossover on the call vector [62]. A single site is randomly chosen for the facility vector, and the parent chromosomes are exchanged at this juncture to produce facility vectors for the offspring chromosome. By reducing the convergence rate, the transformation operator in MA is utilized to avert convergence to local optima. The facility and call vectors are mutated using two separate operators in our model. To create the facility vector, we take all the existing station assignments and swap out one site at random for a facility that we choose from all the available, unallocated stop. The solution's variety is enhanced since this allows for the adoption of allocation of new facility. Order to generate the call vector, we take a chromosomal element and replace its current value with a random value that falls within its allowable range. By exploring the area around a given solution, a local search (Iterated Simulated Annealing with Randomized Tabu Search) was employed to find the best solution locally. This technique involves relocating an ambulance (ALS/BLS) from node i to an adjacent node j , where node j regarded as the optimal position within the neighbourhood area. All move pairs (i, j) are stored in the long-term memory provided that the tabu list remains effective. The extent of the tabu list is expanded to include an action if it recurs within the long-term memory. Meanwhile, if the identical move does not recur within $(2K_t^A, 2K_t^B)$ iterations for ALS/BLS, respectively, it is removed from the tabu list [42]. The initial ambulance is assigned to the primary operation, with the most suitable node from its solely if the pair is not present on the tabu list. The regions covered by the ALS/BLS ambulances within the proposed search algorithm are delineated as the Moore Neighbourhood. For additional further details regarding the Moore Neighbourhood, readers are referred to [63]. Solution fitness evaluation follows priority as: compute Z_1 then reject solutions that violating constraint (1) and compute Z_2 only if feasible. Comparison rule: priorities higher Z_1 and break ties using Z_2 while keeping Z_1 optimal. A predefined group of offspring is produced during each iteration to form a reservoir of offspring by repeated picking a pair of parents from the premier population. The offspring replace the least effective solution within the current generation. The entire procedure is carried out until a specified number of iterations or convergence is attained.

5 Computational experiments

5.1 Study area and data description

The computational experiments are based on emergency medical service demand data obtained from Al-Yarmouk Hospital in the Al-Karkh district of Baghdad, Iraq. The district has an estimated population of approximately 740,684 inhabitants and covers an area of about 477,834 square miles [17]. The EMS system in this region operates using two kinds of ambulances: ALS that equipped to handle life threatening emergencies and BLS that primarily used for non-critical medical cases [17]. A dataset containing 7205 emergency calls was used for the experiments. These calls were classified into two groups: critical (life threatening) and non-critical. To represent the spatial distribution of demand, the study area was divided into 2×2 mile square node grid network, producing 144 demand nodes across Al-Karkh district [42]. To capture the temporal variability of EMS demand, a day was divided into three planning periods: 12:00 am to 8:00 am (off-peak demand), 8:00 am to 4:00 pm (peak demand), and 4:00 pm to 12:00 am (moderate demand). This temporal segmentation allows the model to account for fluctuations in call arrivals throughout the day. In essence, the user establishes the proper length of each time period, which may be determined by statistical analysis of the data or by how the intended experiment is carried out. The EMS system must satisfy specific response time requirements depending on the severity of call. Thus, critical calls must be served by either ALS or BLS within 8 min and non-critical calls must be served by BLS ambulance within 10 min. For both call types, a minimum reliability level of 0.90 is required.

The proposed model was solved using MA-ISA algorithm that implemented in C ++, and all experiments were executed on a computer with specifications: Intel Core (TM) i7-1165G7 (2.80 GHz), RAM 16 GB. Tables 1 and 2 below present the specifics of the issue parameters and MA-ISA algorithm parameters that used in this part.

Table 1 Model parameters

Total critical calls	5052
Total non-critical calls	2135
Number of ALS ambulances available	30
Number of BLS ambulances available	15
Minimum reliability of ALS/BLS ambulances	0.90
Minimum coverage requirement of critical/non-critical	0.95
Time standard for ALS ambulances (TH^A) (minute)	8
Time standard for BLS ambulances (TH^B) (minute)	10
Mean service time of ALS ambulances (minute)	0.02
Mean service time of BLS ambulances (minute)	0.025

Table 2 MA-ISA parameters

Crossover probability	0.90
Mutation probability	0.05
Population size	100
Maximum iterations	1000

5.2 Comparison with exact solver and alternative meta-heuristics

To evaluate the equality of the solutions produced by the proposed MA-ISA algorithm, a set of benchmark experiments was conducted using small scale EMS instances. For these benchmark instances, the stochastic EMS survival model was solved using two different approaches:

1. Exact solution method implemented as a mixed integer nonlinear programming MINLP formulation solved using IBM ILOG CPLEX optimization studio.
2. Alternative meta-heuristic algorithm derived from the dynamic coverage model proposed by Naji et al. [41] that used a stochastic travel time framework similar to the one adopted in the suggested model and therefore provides a suitable baseline for comparison.

The inclusion of these two methods allows the proposed MA-ISA algorithm to be evaluated against both an exact optimization approach and state-of-the-art heuristic framework used in EMS location allocation models. Because the proposed EMS location model incorporates several nonlinear components including survival probability integration, stochastic travel time modeling and hypercube based queue approximations, the exact solver becomes computationally expensive for longer problem instances. Consequently, the exact solver was used only for small instances where optimal solutions could still be obtained within reasonable time limits. Three algorithms were therefore compared: exact MINLP model solved by CPLEX, the heuristic algorithm for dynamic coverage model Naji et al. [41], and the proposed MA-ISA algorithm. Table 3 shows the benchmark comparison results.

The results present in Table 3 demonstrate that the proposed MA-ISA algorithm is capable of producing solutions that are very close to the optimal solutions obtained by the exact MINLP solver. In particular, the difference between the MA-ISA solutions and optimal solutions obtained by CPLEX remains within 1–2% for both survival probability and coverage objectives. However, the computational efficiency of the MA-ISA is substantially higher. While the exact solver required several minutes to solve even small instances the MA-ISA produced high quality solutions within a few seconds. Furthermore, when compared with Naji algorithm, the proposed MA-ISA approach consistently achieves higher survival probabilities and better coverage level. This improvement can be attributed to the incorporation of survival probability and hypercube based ambulances availability estimation, which provide a more realistic representation of the EMS operations under stochastic demand conditions.

Table 3 Benchmark comparison results

Instance	Nodes	Facilities	Method	Survival objective for critical calls	Coverage objective for non-critical calls	CPU time (per seconds)
S1	10	4	CPLEX	0.892	0.812	210
			Naji algorithm	0.872	0.785	18
			MA-ISA	0.889	0.814	14
S2	15	6	CPLEX	0.904	0.833	420
			Naji algorithm	0.891	0.800	27
			MA-ISA	0.904	0.827	21

5.3 Scalability analysis using simulated instances

Although the real dataset from the Al-Karkh District provides valuable insights into the performance of the proposed model, evaluating the algorithm using only one dataset may limit the generalizability of the results. To address this limitation, additional simulated EMS networks of varying sizes were generated. These synthetic datasets were designed to represent different urban EMS environments with varying numbers of demand zones and candidate ambulance stations. Simulated instance characteristics presents in Table 4 below.

The simulated datasets were generated according to commonly used stochastic EMS modeling assumptions:

- Emergency call arrivals were generated using Poisson process, which is widely used to model emergency incident arrivals in EMS system.
- Travel time were modeled using a Weibull distribution, which captures the variability observed in real urban travel time data.

For each instance size, 20 independents simulation scenarios were generated. The propose MA-ISA was then applied to each scenario, and the average performance metrics were recorded in Table 5.

Table 4 Instance characteristics

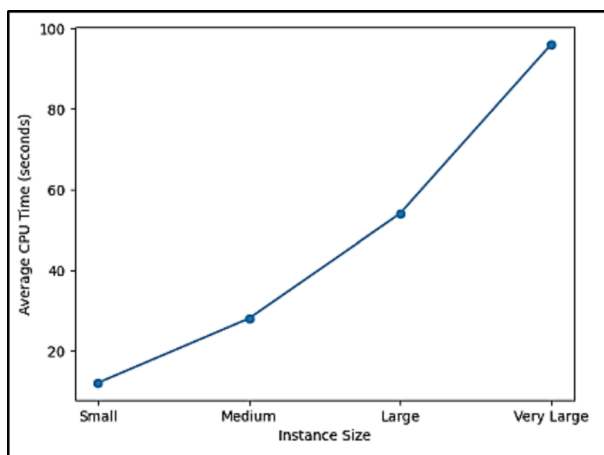
Instance size	Demand nodes	Candidate stations	Time periods
Small	10	4	3
Medium	25	8	3
Large	50	12	3
Very large	100	20	3

Table 5 Scalability results

Instance size	Average of survival	Average of coverage	CPU time (per seconds)
Small	0.897	0.814	12
Medium	0.902	0.844	28
Large	0.911	0.867	54
Very large	0.919	0.885	96

The results in the Table 5 indicate that the proposed MA-ISA algorithm scales efficiently as the problem size increases. As the number of demand calls and candidate stations grows, the algorithm continues to generate high quality solutions while maintaining reasonable computational times. Even for the largest simulated instance containing 100 demand nodes and 20 candidate stations, the MA-ISA algorithm was able to produce quality solutions within approximately 1–2 min. This indicates that algorithm remains computationally tractable for realistic EMS planning scenarios. Figure 1 illustrates the computational scalability of MA-ISA as the size of the EMS network increases.

On the other hand, Fig. 2 illustrate the convergence behavior of the MA-ISA during optimization process. This convergence pattern demonstrates that the integration of MA framework with ISA local search procedure effectively balances exploration and exploitation, allowing the algorithm to converge to near optimal solutions within a relatively small number iteration.

**Fig. 1** Scalability of MA-ISA Algorithm

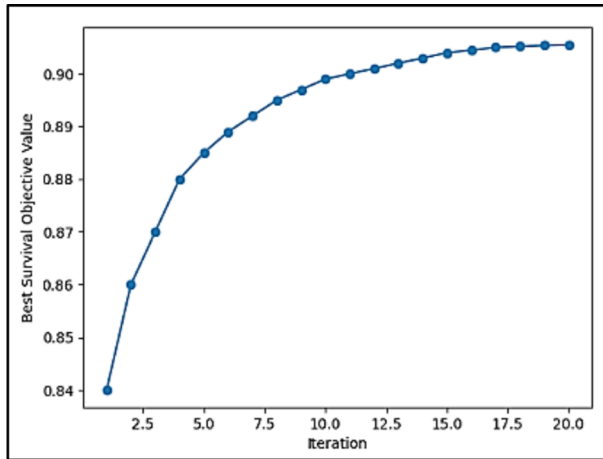


Fig. 2 Convergence Behavior of MA-ISA Algorithm

5.4 Computational efficiency and model scope

Although the computational time required to solve the proposed model is relatively small, the model itself isn't intended for real time dispatch decision making. Instead, the proposed framework is designed as a strategic and tactical planning tool for EMS administrators. More specifically, the model determines the optimal deployment of ALS and BLS ambulances across predefined planning periods based on historical emergency patterns. These deployment plans are computed offline prior to daily EMS operations. Once the optimization results are obtained, the recommended ambulances locations and fleet allocations can be implemented during the operational period. Therefore, the computational efficiency of propose MA-ISA primarily supports rapid scenario analysis and strategic planning, allowing EMS planners to evaluate multiple demand scenario, fleet size and coverage policies within a short period time.

6 Results and discussion

Dataset from Al-Yarmouk Hospital, located in the Al-Karkh District, Baghdad, Iraq, is used in this part to investigate the application of the suggested model inside the EMS system. The optimal solutions obtained by the proposed MA-ISA algorithm for each time period are presented in Tables 6 and 7.

The results from Table 6 indicate that survival rates remain consistently high across all periods, with the highest value observed during the off-peak period. This improvement is primarily due to a reduction in emergency demand, which lowers ambulance busy probabilities and increase the likelihood that an available unit can respond within the target response time. Furthermore, Table 7 indicates that BLS ambulances deliver the highest cover percentage for non-critical demand at the initial time period, which

Table 6 Geographical distribution of ALS/BLS ambulances, average percentage of survival of critical demand, and CPU run based on the specified time period

Time period	Geographical distribution		Average percentage of survival for critical calls	CPU run time (minutes: seconds: milliseconds)
	ALS ambulances	BLS ambulance		
12:00 am – 8:00 am	33, 42, 67, 83, 95, 101, 122	62, 64, 99	0.97079	1:37.175
8:00 am – 4:00 pm	28, 39, 52, 54, 57, 74, 80, 87, 90, 111, 124, 135	37, 57, 66, 78, 93	0.96304	2:35.112
4:00 pm – 12:00 am	21, 43, 52, 81, 102, 112, 121, 137	53, 78, 90, 122	0.98126	2:03.54

Table 7 Geographical distribution of BLS ambulances, average cover percentage of non-critical demand, and CPU run based on the specified time period

Time period	Geographical distribution	Average cover percentage of non-critical calls	CPU run time (minutes: seconds: milliseconds)
12:00 am – 8:00 am	28, 43, 112, 100, 101, 128	0.96127	1:09.11
8:00 am – 4:00 pm	22, 46, 62, 67, 76, 87, 97, 111, 136	0.95021	2:33.29
4:00 pm – 12:00 am	12, 38, 67, 83, 98, 113, 127	0.95897	1:08.49

aligns with the observation that both categories of demand calls are unevenly distributed throughout the Al-Karkh district during this timeframe. The coverage rate remains above 95% for all scenarios, demonstrating the robustness of the proposed model. Figures 3 show the survival probability for critical calls, while Figs. 4 shows coverage vs. reliability level of non-critical calls over period times. In contrast, the runtime increases during the second time period that corresponds to the peak demand interval. During this period, the algorithm must evaluate a larger number of feasible ambulance deployment configurations, resulting in increased computational effort.

6.1 Sensitivity analysis with respect ambulances reliability

To further evaluate the robustness of the suggested EMS model, a sensitivity analysis was conducted to examine the influence of ambulance reliability on the performance of the system. In particular, the experiments analyze how different reliability level affect the survival probability of critical calls and the coverage level of non-critical calls. This enables decision-makers to make suitable choices when using proposed approach in relevant contexts. To investigate this effect, several computational experiments were

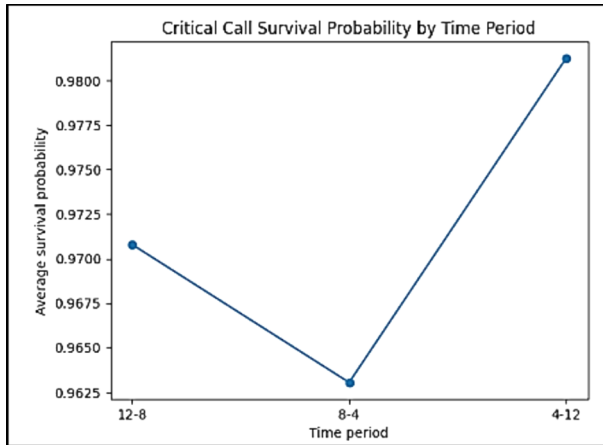
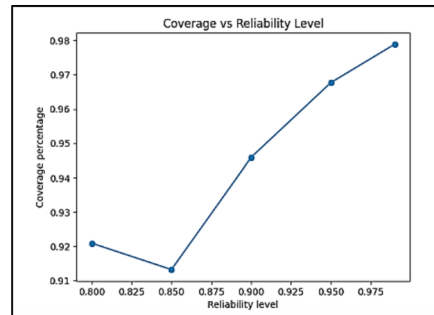


Fig. 3 Survival probability of critical calls

Fig. 4 Coverage vs. reliability level of non-critical



performed using reliability levels of 0.8, 0.85, 0.9, 0.95 and 0.99. for each reliability level, the proposed model was solved using MA-ISA algorithm while keeping all other model parameters constant. The resulting ambulance deployment configurations, corresponding survival probabilities of critical calls and coverage percentages of non-critical calls, are summarized in Table 8.

The results presented in Table 8 demonstrate that the system performance improves as the reliability level of ambulances increases. In particular, higher reliability values lead to higher survival probabilities for critical demand and improved coverage rates for non-critical demand. The average survival probabilities increase significantly across all planning periods when the reliability level increases. This improvement occurs because higher reliability reduces the likelihood that ambulances become unavailable due to failures or operational constraints, thereby increasing the probability that an emergence call be served within the required response time threshold. Similarly, coverage percentages for non-critical calls also improve as reliability increases. When ambulance reliability is low, ambulances are more likely to be unavailable due to

Table 8 Geographical distribution of ALS/BLS ambulances, average percentage of survival, average cover percentage under different reliabilities for Al-Karkh district

Reliability of servers	Time period	Locations of ALS/BLS ambulances to cover the critical calls		Average of survival	Locations of BLS ambulances to cover the non-critical calls	Average of coverage
		ALS ambulances	BLS ambulances			
0.80	12:00 am – 8:00 am	(14, 79, 130)	(12, 63)	0.93001	(36, 55, 90)	0.9399
	8:00 am – 4:00 pm	(37, 74, 88, 123)	(8, 57, 88)	0.9263	(22, 56, 113, 130)	0.9187
	4:00 pm – 12:00 am	(74, 87, 112)	(13, 44)	0.9345	(29, 57, 93, 136)	0.9219
	12:00 am – 8:00 am	(28, 48, 55, 124)	(92)	0.94112	(30, 53, 90)	0.9204
0.85	8:00 am – 4:00 pm	(12, 23, 43, 46, 112)	(66, 87)	0.92408	(54, 77, 101, 105, 130)	0.91765
	4:00 pm – 12:00 am	(8, 43, 54, 77, 87)	(23, 124)	0.93914	(22, 38, 65, 136)	0.910981
	12:00 am – 8:00 am	(36, 48, 76, 85, 93, 111, 132)	(62, 64)	0.9761	(38, 43, 100, 102, 112, 130)	0.9556
	8:00 am – 4:00 pm	(28, 39, 52, 54, 57, 74, 80, 87, 90, 111, 114, 130)	(37, 57, 78)	0.9598	(22, 46, 62, 67, 76, 91, 98, 112, 130)	0.9400
0.95	4:00 pm – 12:00 am	(21, 40, 55, 75, 100, 102, 111, 114, 136)	(53, 78, 122)	0.9677	(35, 38, 57, 73, 78, 80, 131)	0.94321
	12:00 am – 8:00 am	(8, 29, 32, 44, 56, 74, 93, 113, 130)	(67, 75, 80)	0.97991	(14, 36, 48, 55, 64, 87, 93, 123, 136)	0.9576
	8:00 am – 4:00 pm	(31, 36, 39, 46, 52, 57, 62, 64, 67, 81, 88, 98, 94, 125, 129)	(20, 24, 73, 102, 106, 111)	0.9623	(8, 13, 22, 35, 40, 52, 55, 58, 73, 76, 82, 97, 100, 103, 106, 135, 141)	0.9499
	4:00 pm – 12:00 am	(8, 21, 44, 55, 62, 67, 77, 88, 97, 102, 123, 125, 131, 136)	(27, 39, 99, 105, 140)	0.96899	(14, 44, 56, 64, 77, 80, 88, 93, 99, 100, 110, 114, 130, 136)	0.9511

Table 8 (continued)

Reliability of servers	Time period	Locations of ALS/BLS ambulances to cover the critical calls		Average of survival	Locations of BLS ambulances to cover the non-critical calls	Average of coverage
		ALS ambulances	BLS ambulances			
0.99	12:00 am – 8:00 am	(5, 43, 55, 61, 67, 76, 87, 99, 99, 102, 123, 124, 131, 141)	(20, 24, 39, 103, 136)	0.9856	(8, 14, 21, 23, 44, 56, 63, 77, 80, 88, 93, 100, 110, 114, 130, 135, 140)	0.9500
	8:00 am – 4:00 pm	(7, 20, 24, 35, 37, 44, 46, 49, 56, 58, 64, 67, 73, 79, 86, 83, 99, 103, 112, 129, 132, 136)	(15, 39, 58, 62, 90, 110, 131, 140)	0.9775	(5, 14, 21, 28, 31, 43, 46, 53, 58, 62, 67, 77, 81, 86, 97, 100, 110, 105, 122, 125, 131, 136, 140)	0.97229
	4:00 pm – 12:00 am	(5, 15, 43, 46, 53, 55, 58, 62, 67, 73, 75, 77, 81, 97, 100, 105, 110, 112, 122, 136, 140)	(21, 24, 31, 35, 86, 93, 125, 131)	0.9881	(8, 13, 23, 37, 39, 49, 52, 56, 58, 73, 79, 83, 93, 99, 110, 103, 106, 135, 131, 140)	0.9501

busy states. Another important observation from Table 8 is that higher reliability levels typically require the deployment of additional ambulances across the candidate stations. Increasing the number of ambulances improves the likelihood that at least one ambulance remains available even when others are already responding to ongoing incidents. This redundancy helps maintain service performance under stochastic demand conditions. The proposed approach enables decision-makers (DM) to identify the ideal locations for ALS/BLS units stations and allocation of units to each station. Furthermore, DM may ascertain the preferred hierarchy of stations inside each zone, namely the ranking of stations. Finally, the proposed model enables DM to optimize an allocation of available ambulances to enhance cover and survival probabilities for various patient categories. The computational experiments demonstrate that suggested MA-ISA algorithm is capable of producing high quality solutions within short computational times, making it a practical tool for EMS planning and scenario analysis.

7 Conclusion

This study introduces a stochastic optimization EMS location model that integrates two categories of ambulances: ALS for critical demand and BLS for non-critical demand. The BLS ambulances are designated for two responsibilities: providing "back-up" support for important calls and covering non-essential calls. Suggested model jointly considers two categories of demand (critical and non-critical) and aims to maximize the survival probability of critical calls while maintaining adequate coverage of non-critical calls. The model incorporates several key features that enhance the realism of EMS planning models. First, the stochastic nature of response times as the travel times are modeled using a Weibull distribution. Second, patient outcomes for critical calls are represented through a logistic survival function, allowing the model to explicitly account for the relationship between response time and survival probability. Third, ambulance availability is estimated based on an approximation hypercube model that capture sever dependencies and busy probability of ambulances. To reflect operational priorities in emergency care system, the model adopts a priority optimization framework in which survival probability for critical calls is maximized before improving the coverage for non-critical calls. Because the resulting problem is nonlinear, stochastic and combinatorial, an exact solution approach becomes computationally impractical for realistic EMS system. therefore, a hybrid meta-heuristic solution method based on Memetic algorithm combined with an ISA local search procedure (MA-ISA) was developed. MA-ISA algorithm integrates global exploration through population-based search with intensive local improvement, enabling the efficient identification of quality deployment strategies. The proposed model and algorithm were evaluated through a comprehensive set of computational experiments based on EMS demand patterns derived Al-Karkh district of Baghdad. In addition to the original case study data, several simulated EMS networks of varying sizes were generated order to evaluate the robustness and scalability of the suggested algorithm. Furthermore, benchmark experiments on small instances were conducted to compare the performance of the MA-ISA algorithm with solutions obtained using an exact optimization solver as well as with an alternative meta-heuristic approach from dynamic coverage model

[41]. The results show that the proposed algorithm consistently produces solutions that are very close to the optimal benchmark solution while requiring lower CPU times. Computational experiments also highlight important insights for EMS planning. The results demonstrate that explicitly incorporating survival probability into the objective function substantially improves outcomes for critical demand compared with approaches that focus solely on coverage metrics. In addition, the integration of ALS and BLS ambulance types provides greater flexibility in responding to stochastic demand patterns, while the hypercube based availability approximation allows the model to realistically capture ambulance busy probabilities without excessive computational complexity. Sensitivity analyses further show that ambulance reliability levels and demand fluctuations effect system performance, particularly during peak periods. It is important to note that the proposed model is designed as a strategic tactical planning tool rather than a real time dispatch model. The model determines optimal ambulance deployment plans for predefined planning periods based on historical demand patterns. These deployment decisions are computed offline prior to daily operations and provide guidance for EMS planners regarding station locations and fleet allocation across different time periods.

Despite the contributions of this research, several directions for future work remain. The effect of seasonal demand changes, which may have a considerable impact on ambulance placement, is not taken into consideration in this research [64]. When it comes to emergency management system design, equity is an essential factor to take into account since fluctuations in demand might result in an uneven allocation of resources. There is a compelling route for research that may be pursued by combining coverage and survival function-based goals with an equity-based goal function [65]. The development of multi-objective models that make use of a variety of performance indicators and solution approaches in order to produce Pareto optimum solutions is a possible path for future study [62, 66]. The design of experiments methodology may be used to optimize the parameters of the suggested solution strategy. A potential avenue for future study is devising an exact solution method that leverages the problem's structure to enhance solution quality.

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Data availability The data used and analyzed during the current study were obtained from Al-Yarmouk hospital, Al-Karkh district, Baghdad, Iraq. The datasets are available from the author upon reasonable request.

Declarations

Conflict of interests The author declares that there are no competing interests relevant to the content of this manuscript.

Ethics approval This study did not involve human participants or animals, and therefore ethical approval was not required.

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